

Behavior of Foodborne Pathogens *Listeria monocytogenes* and *Staphylococcus aureus* in Mixed-Species Biofilms Exposed to Biocides

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ABSTRACT In nature and man-made environments, microorganisms reside in mixed-species biofilms, in which the growth and metabolism of an organism are different from these behaviors in single-species biofilms. Pathogenic microorganisms may be protected against adverse treatments in mixed-species biofilms, leading to health risk for humans. Here, we developed two mixed five-species biofilms that included one or the other of the foodborne pathogens *Listeria monocytogenes* and *Staphylococcus aureus*. The five species, including the pathogen, were isolated from a single food-processing environmental sample, thus mimicking the environmental community. In mature mixed five-species biofilms on stainless steel, the two pathogens remained at a constant level of $\sim 10^5$ CFU/cm². The mixed five-species biofilms as well as the pathogens in mono-species biofilms were exposed to biocides to determine any pathogen-protective effect of the mixed biofilm. Both pathogens and their associate microbial communities were reduced by peracetic acid treatments. *S. aureus* decreased by 4.6 log cycles in monospecies biofilms, but the pathogen was protected in the five-species biofilm and decreased by only 1.1 log cycles. Sessile cells of *L. monocytogenes* were affected to the same extent when in a monobiofilm or as a member of the mixed-species biofilm, decreasing by 3 log cycles when exposed to 0.0375% peracetic acid. When the pathogen was exchanged in each associated microbial community, *S. aureus* was eradicated, while there was no significant effect of the biocide on *L. monocytogenes* or the mixed community. This indicates that particular members or associations in the community offered the protective effect. Further studies are needed to clarify the mechanisms of biocide protection and to identify the species playing the protective role in microbial communities of biofilms.

IMPORTANCE This study demonstrates that foodborne pathogens can be established in mixed-species biofilms and that this can protect them from biocide action. The protection is not due to specific characteristics of the pathogen, here *S. aureus* and *L. monocytogenes*, but likely caused by specific members or associations in the mixed-species biofilm. Biocide treatment and resistance are a challenge for many industries, and biocide efficacy should be tested on microorganisms growing in biofilms, preferably mixed systems, mimicking the application environment.

KEYWORDS *Listeria monocytogenes*, *Staphylococcus aureus*, pathogen, mixed-species biofilm, processing environment, biocide

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To prevent contamination, infection, or foodborne disease in the clinical or food producing sector, antimicrobials, detergents, and biocides are used to inactivate or eradicate microorganisms (1, 2). Most guidelines for biocides include testing of efficiency on planktonic pure cultures of microorganisms, but little is known about their efficacy on microbial biofilms (3). Moreover, most microorganisms live in complex biofilms (4) composed of multiple species (5, 6). In biofilms, microorganisms can cooperate (7) and protect themselves from adverse environmental conditions. Thus, several studies have reported that sessile cells can be up to 1,000-fold more resistant than cells in a planktonic state (8–10).

Biofilm formation and resistance to biocide treatment have been studied and recognized as important factors that contribute to the survival and persistence of microbial contamination in drinking water (11) and in oral hygiene (12, 13). The food processing environment is also believed to provide conditions for biofilm development, including polymicrobial biofilms. Although many studies have focused on monoculture systems, it is being recognized that biofilms are predominantly polymicrobial (14, 15). The behavior of microorganisms in a mixed biofilm differs from the behavior of a monospecies biofilm (16, 17), and for instance, resistance to antimicrobials can be increased in the mixed system (18). The complete picture on mechanisms involved in resistance has not been fully unraveled, but some of the reasons for biofilm resistance to antimicrobial compounds are proposed to reside in the specific architecture, the decreased metabolic activity, or the presence of extracellular matrix (19). In addition, conditions in a processing environment, such as temperature, have been shown to decrease biocide efficiency on biofilms (20, 21).

A major concern raised by the above-mentioned observations is whether pathogenic microorganisms can be protected in mixed biofilms (22, 23). It has been demonstrated that *Bacillus subtilis*, resistant to peracetic acid exposure, was able to protect *Staphylococcus aureus*, usually sensitive to this disinfectant, in a dual-species biofilm (24). Indeed, biofilms in food processing environments have been shown to contribute to foodborne pathogen survival in cleaning and disinfection treatments, leading to persistence of those microorganisms (25).

Listeria monocytogenes is a ubiquitous microorganism, and it can cause serious foodborne disease. This psychrotrophic bacterium has been found in different food products that can lead to listeriosis after ingestion (26). *L. monocytogenes* attaches to surfaces, but its ability to form biofilms is controversial (27, 28). Nevertheless, clones of *L. monocytogenes* can survive and persist in niches of processing environments for several years despite cleaning and disinfection procedures (29, 30).

S. aureus is one of the most common causative agents of food poisoning and is also involved in nosocomial infections (31). *S. aureus* can form biofilms on different abiotic surfaces found in food processing environments such as glass, stainless steel, polypropylene, and polystyrene (32). The food processing environment can also provide suitable conditions for *S. aureus* biofilm production, which is enhanced by suboptimal growth temperature as well as glucose and sodium chloride availability (33). Furthermore, increasing resistance of *S. aureus* in biofilms toward disinfectants in processing environments has been reported in several studies (21).

Both pathogens are commonly found in food processing environments, including dairy, meat, and seafood worldwide (34, 35), and are reported to be common contaminating agents in the Brazilian dairy industry (36–38). Furthermore, several studies have highlighted that the two pathogens are often found in polymicrobial communities (6, 23, 39, 40).

The effectiveness of biocides depends on the composition of the food soil (i.e., organic matter remaining after insufficient surface cleaning) and the temperature (3) as well as the antimicrobial used and the surface type (41), the treatment exposure time, and the procedure used (10). However, the biofilm mode of growth is also important for effectiveness of biocides. Therefore, the control of biofilm is important for public health in clinical or industrial environments and there is a need for understanding the mechanisms involved in the enhanced pathogen resistance seen in mixed-species

TABLE 1 Bacterial strains used in the present study

Pathogen/community	Background microorganisms	Name of strain	Reference or source
<i>L. monocytogenes</i>		BZ001	42
comLm	<i>Klebsiella</i> spp.	BZ002	This study
comLm	<i>E. coli</i>	BZ003	This study
comLm	<i>Comamonas</i> spp.	BZ004	This study
comLm	<i>Acinetobacter</i> spp.	BZ006	This study
<i>S. aureus</i>		BZ012	43
<i>S. aureus</i>		Sa30	43
comSa	<i>Aeromonas</i> sp.	BZ013	This study
comSa	<i>Lactococcus lactis</i>	BZ014	This study
comSa	<i>Candida tropicalis</i>	BZ017	This study
comSa	<i>Lactobacillus</i> sp.	BZ018	This study

biofilm. To date, most studies have focused on monospecies or dual-species biofilms; however, no studies have investigated biocide efficiency using more-complex biofilm communities.

The purpose of this study was to determine if a “reproducible” mixed-species biofilm model could be established by cocultivating a pathogen with an associated microbial community isolated from the food processing sector. Specifically, we sought to determine if foodborne pathogens, such as *S. aureus* and *L. monocytogenes*, could establish themselves in such a mixed community and how the presence of this more natural scenario affected their sensitivity to commonly used biocides.

RESULTS

Identification of the community members. We isolated *L. monocytogenes* and *S. aureus* from two separate samples in Brazilian dairies (42, 43) and subsequently, from each of these samples, isolated four different microbial strains at random (Table 1). The sample containing *L. monocytogenes* BZ001 (42) also contained *Klebsiella* sp. (BZ002), *Escherichia coli* (BZ003), *Comamonas* sp. (BZ004), and *Acinetobacter* sp. (BZ006). The sample containing *S. aureus* BZ012 (43) also contained *Aeromonas* spp. (BZ013), *Lactococcus lactis* (BZ014), *Candida tropicalis* (BZ017), and *Lactobacillus* sp. (BZ018).

Establishment of the pathogen in a dual-species biofilm. *S. aureus* and *L. monocytogenes* were individually grown in dual-species biofilms with each of their community members (Fig. 1). The total sessile cell counts ranged from 3.5×10^6 to 3.8×10^7 CFU/cm², while *L. monocytogenes* counts ranged between 1.8×10^5 and 1.3×10^7 CFU/cm² in the dual-species biofilms. The total sessile cell count for the *S. aureus* communities ranged from 3.9×10^6 to 9.2×10^7 CFU/cm² and was between 1.0×10^6 and 6.9×10^7 CFU/cm² for the pathogen in the dual-species biofilms.

Stable concentration of the pathogen in mixed five-species biofilm. In the *S. aureus* mixed five-species biofilm, the total sessile count was 1.9×10^6 CFU/cm² and the *S. aureus* sessile cell count was 4.5×10^5 CFU/cm² (Fig. 2). The two different *S. aureus* isolates, BZ012 and Sa30, behaved very similarly in the mixed biofilm (data not shown). The total sessile cell count of the *L. monocytogenes* community was 1.8×10^7 CFU/cm², and the count was 2.3×10^5 CFU/cm² for *L. monocytogenes* sessile cells. All members of the five-species communities remained in the mature biofilm based on the recovery of all members' colony morphology on brain heart infusion (BHI) plates and by PCR detection for the *S. aureus* community (data not shown).

MICs of the biocides. The MIC of peracetic acid for the two pathogens was 0.075% (1/4 of the concentration used in the dairy) when grown as a monoculture. MICs of peracetic acid against the five species grown together were 0.075% for the mixed culture containing *L. monocytogenes* (comLm) and 0.015% (1/20 of the concentration used in the dairy) for the mixed culture containing *S. aureus*. The individual MICs of peracetic acid against the associated microorganisms were 0.075% for the four associated microbial community members of comLm (BZ002, BZ003, BZ004, BZ006), 0.075% for BZ0013, 0.15% for BZ014 and BZ017, and 0.3% for BZ018.

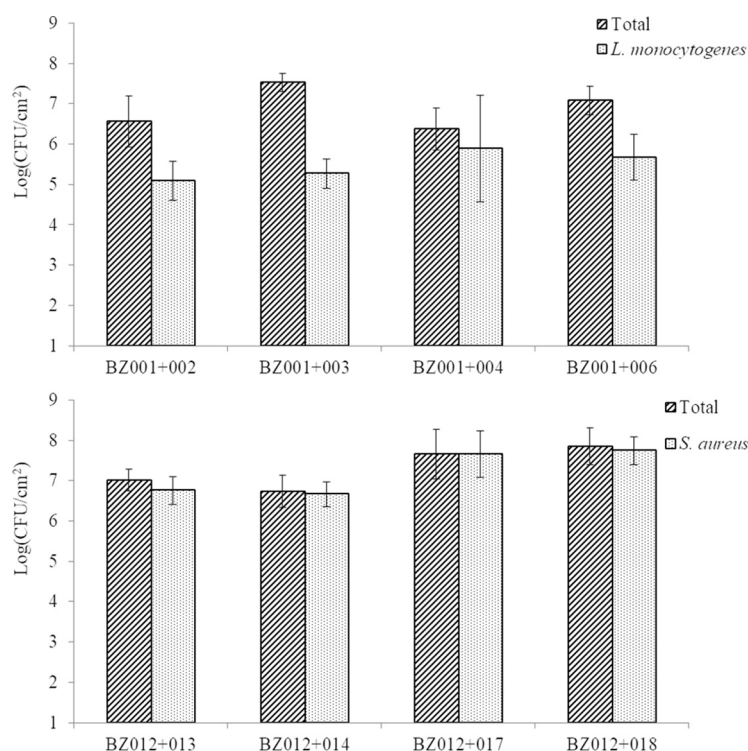


FIG 1 Quantification of viable sessile cells in dual-species biofilms. Each pathogen, *L. monocytogenes* (BZ001) (top) or *S. aureus* (BZ012) (bottom), was grown as a mono- or dual biofilm developed on SSCs with each of the associated community members. At least three biological replicates for *L. monocytogenes* and only duplicates for *S. aureus* were done.

The chlorhexidine digluconate MICs were 0.000390625% for *L. monocytogenes* and 0.0001953125% for *S. aureus*. The MIC of chlorhexidine digluconate against the five species of the *L. monocytogenes*-associated microbial community grown together was 0.003125%. The MIC for chlorhexidine digluconate was not determined for the five-species culture containing *S. aureus* because it was above 0.025%, which was the maximum concentration that could be tested without precipitation.

The full community composition influences biocide susceptibility. Using the monobiofilm and mixed five-species biofilm model described above, biocide susceptibility was assessed. The exposure of a monobiofilm of *S. aureus* to increasing concen-

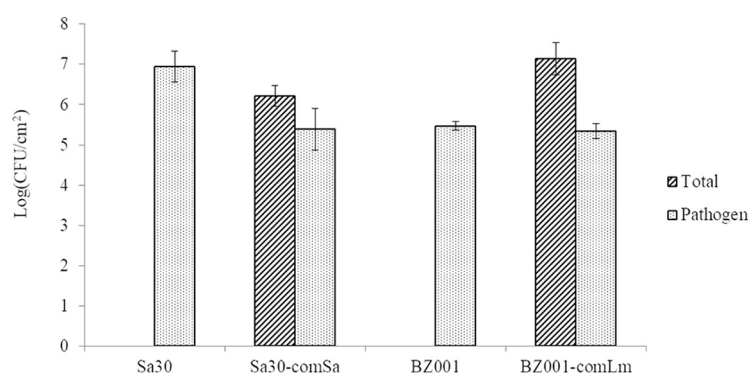


FIG 2 Sessile cell counts of mono- and mixed five-species mature biofilms. Pathogens were *L. monocytogenes* (BZ001) or *S. aureus* (Sa30). The mature biofilms were a 25-h biofilm and a 72-h biofilm for the *S. aureus* community and the *L. monocytogenes* community, respectively. Pathogen cell count was assessed on Baird Parker agar for *S. aureus* and Oxford agar for *L. monocytogenes*. Total community cell count was assessed on BHI agar. Error bars represent the standard deviations for three biological replicates.

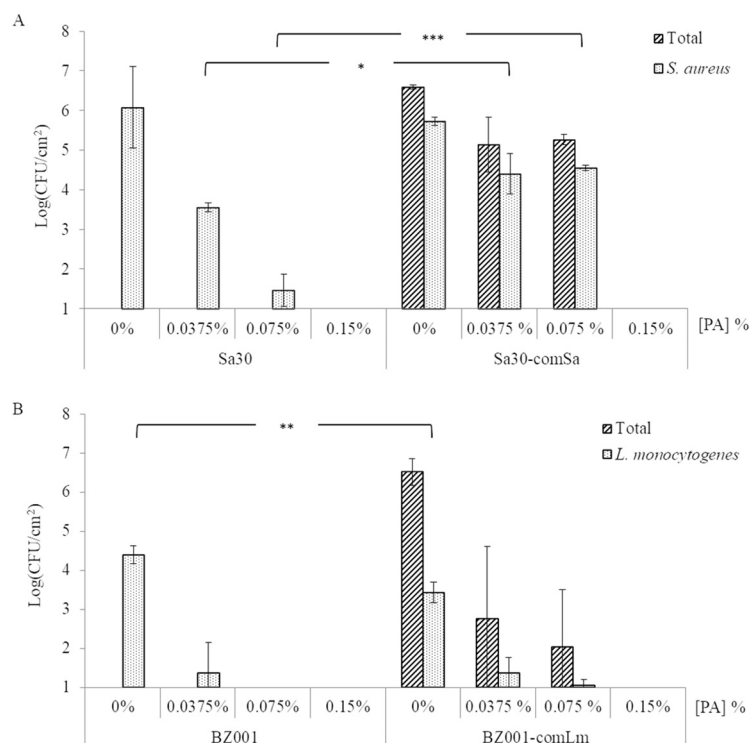


FIG 3 Survival of sessile cells in biofilms formed on SSCs after peracetic acid exposure. For each pathogen, *S. aureus* (Sa30) (A) or *L. monocytogenes* (BZ001) (B), mono- and mixed five-species biofilms were developed on SSCs as mature biofilms before being treated with different concentrations of peracetic acid (PA) for 20 min. Sa30, monobiofilm of *S. aureus*; Sa30-comSa, mixed five-species biofilms containing Sa30; BZ001, monobiofilm of *L. monocytogenes*; BZ001-comLm, mixed five-species biofilms containing BZ001. Results are presented as the means for biological triplicates.

trations of peracetic acid led to a sequential decrease of the *S. aureus* sessile cell survival (Fig. 3A) of 2.5 log between the exposures at 0% and 0.0375% and then a 2.1-log reduction between 0.0375 and 0.075% (MIC value). When *S. aureus* was part of a mixed-community biofilm, 0.075% peracetic acid caused only a 1.1-log reduction of the *S. aureus* sessile cells. This is a significantly ($P = 0.0002$) lower reduction than seen for the monobiofilm (4.6 log). Increasing the peracetic acid concentration from 0.075% to 0.15% fully eradicated *S. aureus* sessile cells from an initial concentration of 4.7×10^6 CFU/cm² in a monobiofilm and 5.5×10^5 CFU/cm² in a mixed five-species biofilm (Fig. 3A). At the same time, no associate microbial community member sessile cells were recovered when the mixed five-species biofilm was exposed to 0.15% peracetic acid (Fig. 3A), meaning that less than 10 CFU/cm² was on the stainless steel coupon (SSC) if not all eradicated.

Treatment of an *L. monocytogenes* monobiofilm with 0.0375% peracetic acid caused a 3-log reduction of the sessile cells (Fig. 3B). The same treatment of the mixed five-species biofilm led to a 2-log reduction of *L. monocytogenes* sessile cells, while the total sessile cells decreased by 3.7 log. When exposed to a higher concentration of peracetic acid, e.g., 0.075%, no viable *L. monocytogenes* sessile cells were recovered from a monobiofilm. The total sessile cells and the *L. monocytogenes* sessile cells decreased by 0.8 log and 0.3 log, respectively, in the mixed five-species biofilm treated with 0.075% peracetic acid compared to treatment with 0.0375% (Fig. 3B). No viable sessile cells were recovered when *L. monocytogenes* was grown as a monospecies biofilm or in a mixed five-species biofilm treated with 0.15% peracetic acid (2-fold higher than the MIC) (Fig. 3B).

No effect of chlorhexidine digluconate was observed. In order to assess susceptibility to other biocides used in the processing environment, the effect of chlorhexi-

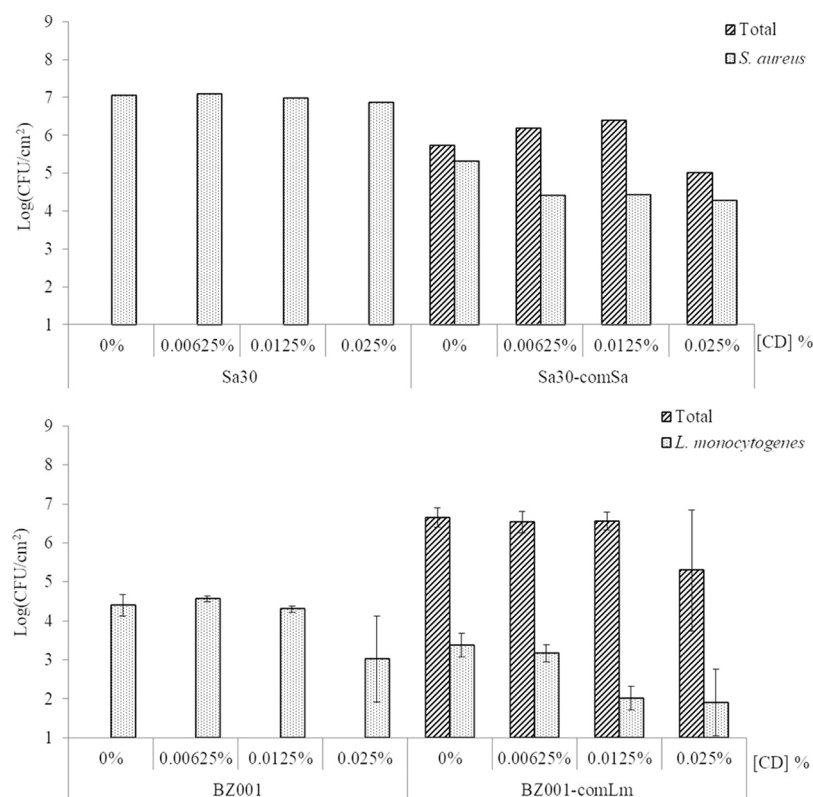


FIG 4 Survival of sessile cells in biofilms formed on SSCs after treatment with different concentrations of chlorhexidine digluconate for 20 min. For each pathogen, *S. aureus* (Sa30) (top) or *L. monocytogenes* (BZ001) (bottom), mono- and mixed five-species biofilms were developed on SSCs as mature biofilms before treatment. Sa30, monobiofilm of *S. aureus*; Sa30-comSa, mixed five-species biofilms containing Sa30; BZ001, monobiofilm of *L. monocytogenes*; BZ001-comLm, mixed five-species biofilms containing BZ001. Results are presented as the means of biological triplicates for *L. monocytogenes* (bottom) and only one biological replicate for *S. aureus* (top).

dine digluconate was also evaluated (Fig. 4). No effect of chlorhexidine treatment was observed on *S. aureus* grown as a monobiofilm or as part of a mixed five-species biofilm (Fig. 4, top).

There was no impact of chlorhexidine digluconate treatment on *L. monocytogenes* for concentrations below 0.0125% (Fig. 4, bottom) whenever the pathogen was grown as a monobiofilm or as part of a mixed five-species biofilm. *L. monocytogenes* sessile cell counts remained steady in a monobiofilm with 10^4 CFU/cm² when treated with a concentration of chlorhexidine digluconate below 0.0125%, while it decreased from 1.6×10^3 CFU/cm² to 1.2×10^2 CFU/cm² in a mixed five-species biofilm (Fig. 4, bottom). This 1.2-log reduction was observed in the monobiofilm when the sessile cells were exposed from 0.0125% to 0.25% chlorhexidine digluconate. The concentration of the total sessile cells, being 4×10^6 CFU/cm², did not change when exposed to 0.00625% and 0.0125% but decreased 1.3 log when exposed to 0.025% of chlorhexidine digluconate. Concentrations higher than 0.025% were not investigated due to technical limitations, as the chlorhexidine digluconate precipitated.

The community rather than the individual influences peracetic acid susceptibility.

To determine if the altered sensitivity to peracetic acid in a mixed-species biofilm was due to the associated microbial community or to the pathogen, we interchanged the pathogens and the communities. When *S. aureus* was grown with the *L. monocytogenes* community members, on the SSC not treated with peracetic acid, the total sessile cells count was 1.3×10^8 CFU/cm² and the *S. aureus* sessile cells count was 2.5×10^4 CFU/cm² (Fig. 5A). No cells were recovered after treatment with 0.075% peracetic acid, indicating that the total sessile cells decreased by at least 6.3 log ($P = 0.001$) and the

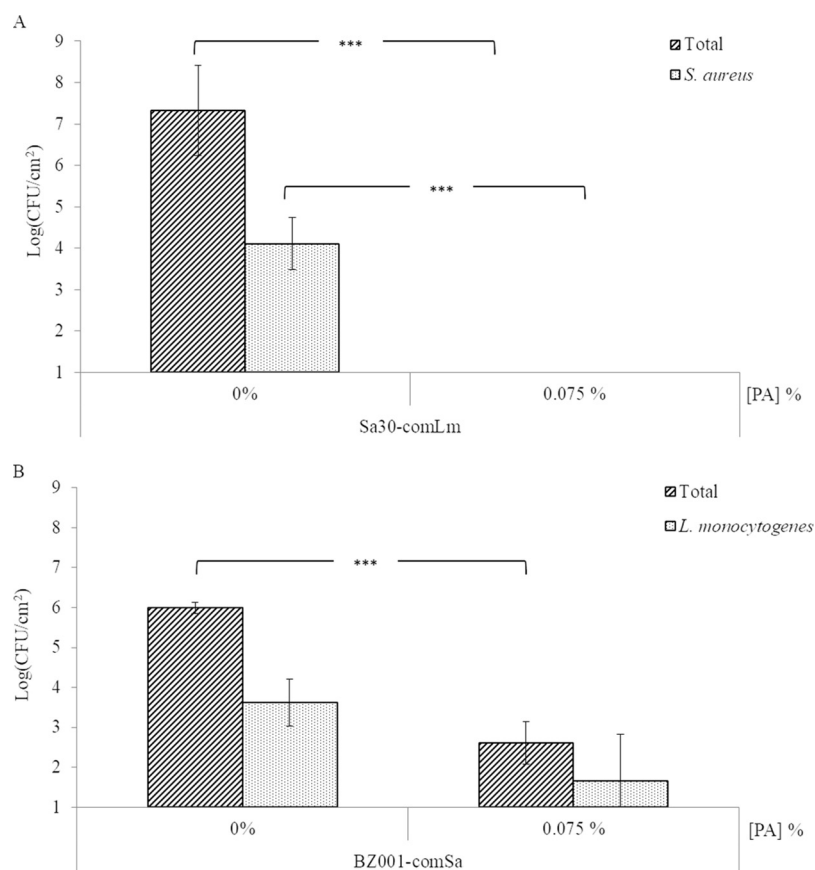


FIG 5 Survival of sessile cells in mixed-species biofilm after 20 min with 0.075% peracetic acid (PA) treatment. Each pathogen, *S. aureus* (Sa30) (A) or *L. monocytogenes* (BZ001) (B), was grown as a mixed five-species biofilm developed on SSCs with the other pathogen community members, i.e., BZ001 associated with comSa and Sa30 associated with comLm. Sa30-comLm, mixed five-species biofilm containing Sa30; BZ001-com, mixed five-species biofilm containing BZ001. Results are presented as the means for biological triplicates.

pathogen sessile cells decreased by at least 3.1 log ($P < 0.001$). When *L. monocytogenes* was grown with the *S. aureus* community members, the overall biofilm production on SSC did not change (Fig. 3B and 5B). Treatment with 0.075% peracetic acid led to a reduction of the total sessile cells by 3.4 log ($P < 0.001$) and a reduction of the *L. monocytogenes* sessile cells by 1.9 log (Fig. 5B).

DISCUSSION

In environments, either natural, clinical, or industrial, microorganisms live mainly in biofilms that are well-structured communities and composed mostly of more than one microbial species (5, 6). However, in a natural or man-made environment, the quantity and diversity of present species raise the complexity underlying the behavior of the biofilms compared to their single-species growth behavior (7). Pathogenic microorganisms can also form biofilms or inhabit biofilms, and this can cause problems for human health, especially if biocide treatment is not fully effective. Also, survival of microorganisms after biocide exposure is one of the results of behavior modification. To date, most biofilm studies focusing on their mechanistic properties or antimicrobial resistance have been done on monospecies biofilms, which do not reflect the complexity reached in a mixed community. Recently, some studies on mixed biofilms have also been conducted, but they have been limited in species diversity (44–47) without focusing on a specific pathogenic organism.

In this study, we set up and studied two mixed five-species biofilms containing a pathogenic bacterium. The ability of some strains to form biofilms fluctuated, with

some dual-species sets of organisms showing better ability to form biofilm than others depending on the strain association. Norwood and Gilmour (48) have shown that *Pseudomonas fragi* and *Staphylococcus xylosus* were the predominant species in biofilms with *L. monocytogenes*, and they described, as have other studies, that some microorganisms can take over in mixed-species biofilms. However, no strains were inhibited in our study. Both in the dual-species and the mixed five-species biofilms, all strains remained in the mature biofilm, including both pathogens. A previous study on mixed biofilms composed of *L. monocytogenes* and *Enterococcus* spp. found that growth at 25°C on stainless steel gave the highest biofilm cell counts (44), supporting the temperature selection that was used in this present study. Furthermore, the mixed five-species biofilm model here was stable and reproducible, leading to a biofilm tool that can be used to study the behavior of a pathogen in mixed biofilms and to unravel the mechanisms involved in survival or persistence despite application of antimicrobial compounds such as biocides.

L. monocytogenes is commonly described as a non-/weak monolayered biofilm producer (23, 49) but can colonize and persist in mixed-species biofilms (23) as also found in the present study. Some authors have described that *L. monocytogenes* was inhibited in biofilms composed of several bacteria compared to monobiofilms (40, 48, 50), e.g., *Staphylococcus sciuri* prevented *L. monocytogenes* from adhering and being part of a biofilm (50). In contrast, the sessile cell counts of *L. monocytogenes* on stainless steel were similar in mono- and mixed five-species biofilms in this study, which is in accordance with other studies (40, 46, 51).

S. aureus, the other major pathogen selected, is a renowned biofilm former (52) and also occurs in multispecies biofilms with *Pseudomonas aeruginosa* (53, 54), *B. subtilis* (24), *Enterococcus faecalis* (55), or *Candida albicans* (53, 56). Some studies have reported that lactic acid bacteria such as *Lactococcus* spp. and *Lactobacillus* spp. (57, 58) can inhibit growth of this pathogen, while that was not noticed in this study. However, the sessile cell counts of each of the associated microbial community members, including the pathogen, decreased in the mixed five-species biofilms compared to the sessile cell counts in mono- or dual-species biofilms, which could be due to competition for nutriment and antimicrobial compound production (58).

Here, dairy sample communities were composed of various microorganisms representing the diversity existing in natural and man-made environments. Several studies have demonstrated that processing environments are composed of a large diversity of microorganisms (59, 60), and Dzieciol et al. (59) have shown that microbial communities were distinct depending on the collecting point, as observed in this study. Hence, it could be suggested that the microbial diversity in processing environments, leading to the biofilm diversity, could impair effective disinfection procedure, as Simões et al. (47) also concluded.

To date, very few studies have investigated antimicrobial resistance of mixed-species biofilms (46, 48) including comparisons with monospecies biofilm data. Using the mature mixed five-species biofilms on stainless steel coupons set up in this study, the biofilm models were subjected to biocides similar to those used in the processing environment where the strains were sampled. The effectiveness of biocide treatment on the two pathogens depended on the biocide used as previously shown (8, 44, 61). Peracetic acid treatment impaired survival in mono- and mixed five-species biofilms, while no significant effect of chlorhexidine digluconate was seen at the tested concentrations. da Silva Fernandes et al. (44) have described peracetic acid as the most efficient biocide used in processing environments compared to quaternary ammonium, sodium hypochlorite, or biguanide, and it was able to eliminate *L. monocytogenes* from the multispecies biofilm as observed in this study at a concentration of 0.15%. Our results indicate that peracetic acid as used in the dairies (0.3%) is efficient in reducing bacterial numbers, since less than 10 CFU/cm² of sessile cells were recovered after treatment with 0.15% peracetic acid (1/2 of the factory in-use concentration). The effect of chlorhexidine digluconate cannot be evaluated since the biocide precipitated at concentrations above 0.025%.

It is noteworthy that sessile cells of *S. aureus* were recovered from both mono- and mixed five-species biofilm (comSa) when they were treated with 0.075% peracetic acid. Also, *S. aureus* sessile cells survived in a monospecies biofilm at this concentration, which was the MIC determined for this pathogen. This corroborates that biofilms can modify and increase the biocide susceptibility response as suggested by some authors (8).

The mixed associated microbial communities, comSa and comLm, protected the pathogens during peracetic acid treatment. Thus, a 1-log difference was obtained between sessile cell counts of *L. monocytogenes* in mono- and mixed five-species biofilms after biocide treatment. Using the same biocide, van der Veen and Abee (62) also observed that *L. monocytogenes* was more resistant to peracetic acid in mixed-species biofilm with *Lactobacillus plantarum* than in monospecies biofilm. In the case of *S. aureus* sessile cells, a decrease of 2.1 log was obtained between treatments at 0.0375% and 0.075% peracetic acid of the monospecies biofilm, while in a mixed five-species biofilm the decrease was only of 0.2 log. Hence, similar to results reported by Bridier et al. (24), *S. aureus* was less susceptible to peracetic acid exposure when the pathogen was part of a mixed-species biofilm than when grown as a monospecies biofilm.

We investigated if the increase of pathogen survival in mixed-species biofilm was due to the specific ability of this pathogen in the associated microbial community or to the community itself. We interchanged the pathogens and the associated microbial communities, i.e., *S. aureus* with comLm and *L. monocytogenes* with comSa. The new mixed five-species biofilms were treated only with 0.075% peracetic acid, which was the highest concentration that allowed survival sessile cells recovery in the first set of experiments. *S. aureus* was established at a lower density in comLm than in the original comSa, while the total sessile cell counts increased. However, *S. aureus* was not protected by the comLm community and no *S. aureus* sessile cells were recovered. In the biofilm composed of *L. monocytogenes* with comSa and treated with 0.075% peracetic acid, a slight increase of survival sessile cells of *L. monocytogenes* was recovered compared to the association with comLm. This could be due to microbial composition variation, comLm being composed of only Gram-negative bacteria while comSa was composed of Gram-positive and -negative bacteria as well as one yeast. Therefore, it is likely the full associated microbial community that is involved in the survival/resistance mechanisms. This protective effect could be due to the microbial species composition, which influences the biofilm matrix composition, as well as the presence of specific microorganisms with higher resistance to the tested biocide; e.g., the MIC for peracetic acid of each member of comLm was 0.075%, whereas members of comSa showed different MIC values ranging from 0.075% to 0.3% (the concentration used in dairy). This is in agreement with the suggestions that biocide resistance could be due to the extracellular matrix or to interactions between the organisms in the mixed biofilm (46, 63).

In conclusion, biofilms can vary due to the microbially diverse populations encountered in man-made environments. This diversity is a key challenge to eradicate unwanted microorganisms. Further studies are required to unravel the exact mechanisms leading to the protective role of the community using complex microbial biofilms mimicking biofilms encountered in industry/clinical or natural environment, as we attempted to set up in this study. Therefore, by improving the knowledge on mixed-species biofilm behavior in the presence of biocides, control of biofilm could be improved in any kind of sector.

MATERIALS AND METHODS

Bacterial strains, growth media, and biocides. The strains of *L. monocytogenes* and *S. aureus* used in this study have been isolated from Brazilian dairies (42, 43) (Table 1). Four strains of the associated microbial community were also isolated as described below. They were isolated from the same sample as the pathogenic bacterium to obtain two communities as encountered in the dairy: one containing *L. monocytogenes* and one containing *S. aureus* (Table 1).

TABLE 2 Primers for species identification

Taxon	Primer	Sequence (5'–3')	Reference or source
Bacteria	27F	AGAGTTTGATCCTGGCTCAG	65
	1492R	GGTTACCTTGTACGACTT	66
Eukaryotes	NL1	GCCATATCAATAAGCGGAGGAAAAAG	67
	LS2	ATTCCTCAACAACTCGACTC	68
<i>S. aureus</i>	16S-Saureus-F	ATACAAAGGGCAGCGAAACC	This study
	16S-Saureus-R	TCATTGTGCCACCTTCGAC	This study
<i>L. lactis</i>	16S-Llactis-F	AAGTTGAGCGCTGAAGGTTG	This study
	16S-Llactis-R	AACGCGGGATCATCTTTG	This study
<i>Lactobacillus</i> spp.	16S-Lbacillus-F	ATTTTGGTCGCCAACGAG	This study
	16S-Lbacillus-R	ACCCACCAACAAGCTAATG	This study
<i>Aeromonas</i> spp.	16S-Aerom-F	GCAGCGGAAAGTAGCTTG	This study
	16S-Aerom-R	ATATCCAATCGCGCAAGG	This study
<i>C. tropicalis</i>	16S-Ctropicalis-F	GTCGAGTTGTTGGGAATGC	This study
	16S-Ctropicalis-R	ATCTCAAGCCCTTCCTTTC	This study

Two isolates of *S. aureus*, BZ012 and Sa30, were used in this study. The isolate BZ012 was selected to represent the sequence type ST398 (43), a major possible health risk ST, and was used for dual-biofilm and mixed-species biofilm setup experiments. The isolate Sa30 was chosen because it represented the major ST/CC trend, ST30/CC1, found in the Brazilian dairy industry (43), and this isolate was used for assessing biocide effects on biofilm containing *S. aureus* (including the MIC assay). It was rationalized that the associated microbial community members isolated from the sample containing *S. aureus* BZ012 would be suitable for any *S. aureus* dairy isolates, and therefore, they were also used as community members with the *S. aureus* Sa30 isolate in the biocide susceptibility experiments.

Isolates were grown on brain heart infusion (BHI) broth (Oxoid), BHI agar (BHI broth, 1.5% agar; AppliChem) or tryptic soy agar (TSA; Oxoid), de Man, Rogosa, and Sharpe agar (MRS), and dichloran rose Bengal chloramphenicol agar (DRBC; Oxoid, UK), and potato dextrose agar plus chloramphenicol (PDA-CAM; Oxoid, UK). *S. aureus* cells were enumerated on Baird Parker agar (Oxoid, UK) with egg yolk emulsion (Oxoid, UK), and *L. monocytogenes* cells were counted on Oxford (Oxoid, UK) with modified *Listeria* selective supplement (Oxoid, UK). Unless otherwise specified, isolates were grown at 37°C and liquid cultures were incubated under shaking conditions at 250 rpm. The isolates were stored in BHI containing 20% (vol/vol) glycerol (Merck, Germany) at –80°C. For biocide treatments, two biocides used in the dairies from where samples were taken were used: peracetic acid (Sigma-Aldrich, Germany) and chlorhexidine digluconate (Sigma-Aldrich, Germany). These stock solutions were diluted in 0.9% (wt/vol) NaCl to the selected concentrations.

Selection of 4 associated microorganisms for each pathogen by phenotypic tests. Using the samples from which *L. monocytogenes* BZ001 (42) and *S. aureus* BZ012 (43) were isolated, four associated community members were isolated by surface plating 0.1 ml of a 10-fold-diluted sample in peptone water (0.1%) supplemented with a 0.85% NaCl (Oxoid, UK) suspension on BHI agar, MRS, and DRBC. MRS and BHI agar plates were incubated at 30°C for 24 to 48 h. DRBC plates were incubated at 25°C up to 7 days and then purified on PDA-CAM. Up to three different colonies of different morphologies were selected and purified on the respective culture media for subsequent analysis. Only one community, i.e., four associated community members, for each pathogen species was selected and used in this study.

Species identification. The isolates were identified by phenotypic characterization and 16S rRNA or 28S rRNA gene sequencing. Shape and motility were determined by microscopy using an Olympus microscope (BX51). Gram reactions were assessed by the 3% KOH method (64). Catalase or cytochrome oxidase was tested using 3% hydrogen peroxide (Merck, Germany) or dry slide (BD Diagnostics, NJ, USA), respectively. DNA manipulation and 16S rRNA gene sequencing using the primer couple 27F (65) and 1492R (66) or NL1 (67) and LS2 (68) for the 28S rRNA gene were performed as described by Oxaran et al. (42) (Table 2).

Development of mature biofilms on SSCs. In order to reproduce the dairy processing environment, biofilm formation was assessed on stainless steel coupons (SSCs) at 25°C (the temperature noted in the investigated dairies) using an associated microbial community of strains isolated from Brazilian dairies (42, 43). Overnight cultures were inoculated to a final optical density at 600 nm (OD_{600}) of 0.01 in a 5-ml BHI broth tube containing a 1- by 2-cm SSC (AISI 316, unpolished, 2B finish, prepared according to Kastbjerg and Gram [69]). Monospecies biofilms were produced from only one single strain, dual-species biofilms were composed of the pathogenic strain (*L. monocytogenes* or *S. aureus*) and one of the associated microbial community members, and mixed five-species biofilms were obtained by the association of a pathogen strain and the four associated microbial community members listed in Table 1 (each inoculated at an equal ratio). Adhesion was performed for 1 h 30 min at 25°C, with shaking at 90 rpm. Subsequently, the medium was gently discarded to remove planktonic and loosely attached cells and replaced with BHI broth followed by incubation at 25°C, with shaking at 90 rpm. To obtain a mature biofilm, the *S. aureus*-containing biofilm was incubated for 25 h and the *L. monocytogenes*-containing biofilm for 72 h.

Enumeration of planktonic and sessile cells. Viable sessile cells in the mature biofilm on SSCs were enumerated by determination of CFU. The SSCs were gently washed three times with 2 ml NaCl (0.9%)

and subsequently sonicated in 2 ml NaCl (0.9%) for 2 min (Delta 220T; Aerosec Industrie). The CFU count of viable sessile cells was assessed by the drop plating method on BHI agar for the total cell count (including all the associated microbial community members), and for the model pathogen cell count, a selective medium was used, i.e., BP or Oxford for *S. aureus* or *L. monocytogenes*, respectively. As growth control, the planktonic cell counts were done using the same method. The results considered the detection limit of the method used in this study, which was 10 CFU/cm².

Species presence assessment. To check that all associated microbial community members remained in the mature biofilm, two methods were used. For the community containing *L. monocytogenes*, species presence was evaluated by colony morphology on BHI agar, as the five community members showed distinct colony morphology. For the community containing *S. aureus*, colonies were not distinguishable on agar plate, and PCR using species-specific primers (Table 2) was used to detect the different organisms. Genomic DNA (gDNA) was extracted using the NucleoSpin Tissue kit (Macherey-Nagel, Germany) from the sessile cells using the same suspension as that used for enumeration, as per the manufacturer's recommendations. PCRs were performed using the TEMPase Hot Start 2× Master mix Blue II (Ampliqon, Denmark) according to the manufacturer's protocol. One microliter of gDNA was added as a matrix. The PCR was run in a Veriti thermal cycler (96-well model 9902; Applied Biosystems), and the amplicon presence was checked by gel electrophoresis.

MIC of biocides. A first assessment of biocide susceptibility was performed using MIC assays in order to target concentrations to be used in the biofilm model. The concentrations used in the dairies were 0.3% and 0.04% for peracetic acid and chlorhexidine digluconate, respectively (S. H. I. Lee, personal communication). Each pathogen (*L. monocytogenes* and *S. aureus* Sa30) was assessed by MIC assay as monoculture and mixed culture (with the four associated microorganisms). The MIC assay was also done for the individual community members by following the same protocol. Overnight cultures were adjusted to an OD₆₀₀ of 0.02 in BHI, and 100 μl was dispensed into wells (96-well, round-bottom microtiter plate; Thermo Scientific, MA, USA) containing 100 μl of a serial dilution of each biocide. The final biocide concentrations tested started at 4.8% and decremented by 2 up to 13 times for peracetic acid and started at 0.05% and decremented by 2 up to 9 times for chlorhexidine digluconate. The MIC was determined after incubation for 24 h at 25°C with shaking at 90 rpm and was defined as the minimum concentration inhibiting growth.

Biocide treatment of mature biofilm on stainless steel coupons. After development of mature mono- or mixed five-species biofilms, sessile cells were subjected to disinfection treatment. SSCs were washed two times in 2 ml 0.9% NaCl-containing plates (6-well plate; Nunc, Denmark) and then immersed in 2 ml 0.9% NaCl solution at the selected biocide concentrations. Biocide treatment was stopped after 20 min of exposure (69) at 25°C with shaking at 90 rpm by transferring the SSCs to 2 ml of Dey-Engley neutralizing broth. Then, the SSCs were sonicated for 2 min in the neutralizing solution and enumeration was performed as described above.

Statistic data analysis. Statistical analyses were performed using the Excel Data Tool to process all data by analysis of variance (ANOVA). Significance was defined with a Fisher test value as a *P* value of ≤0.05.

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We declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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